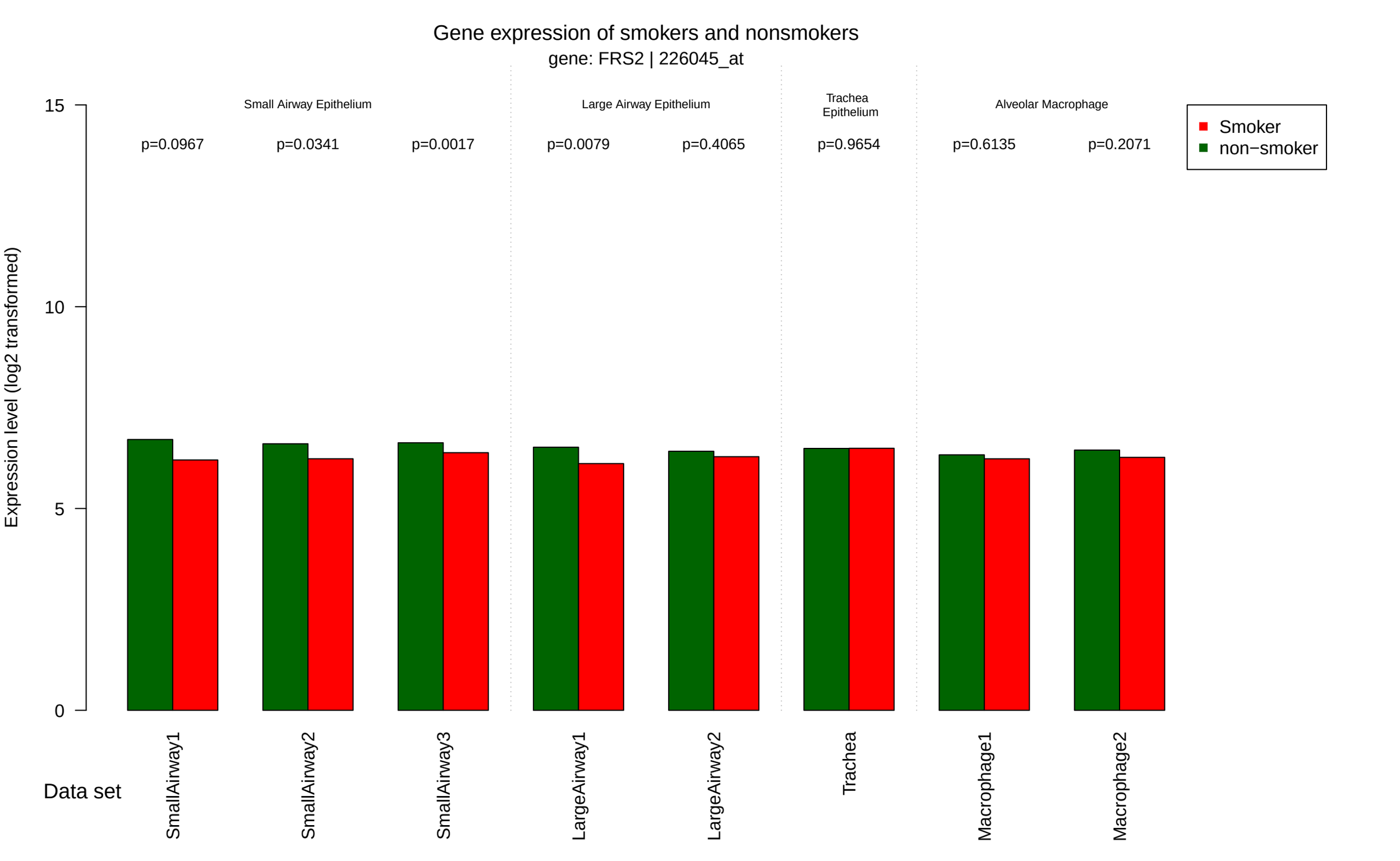
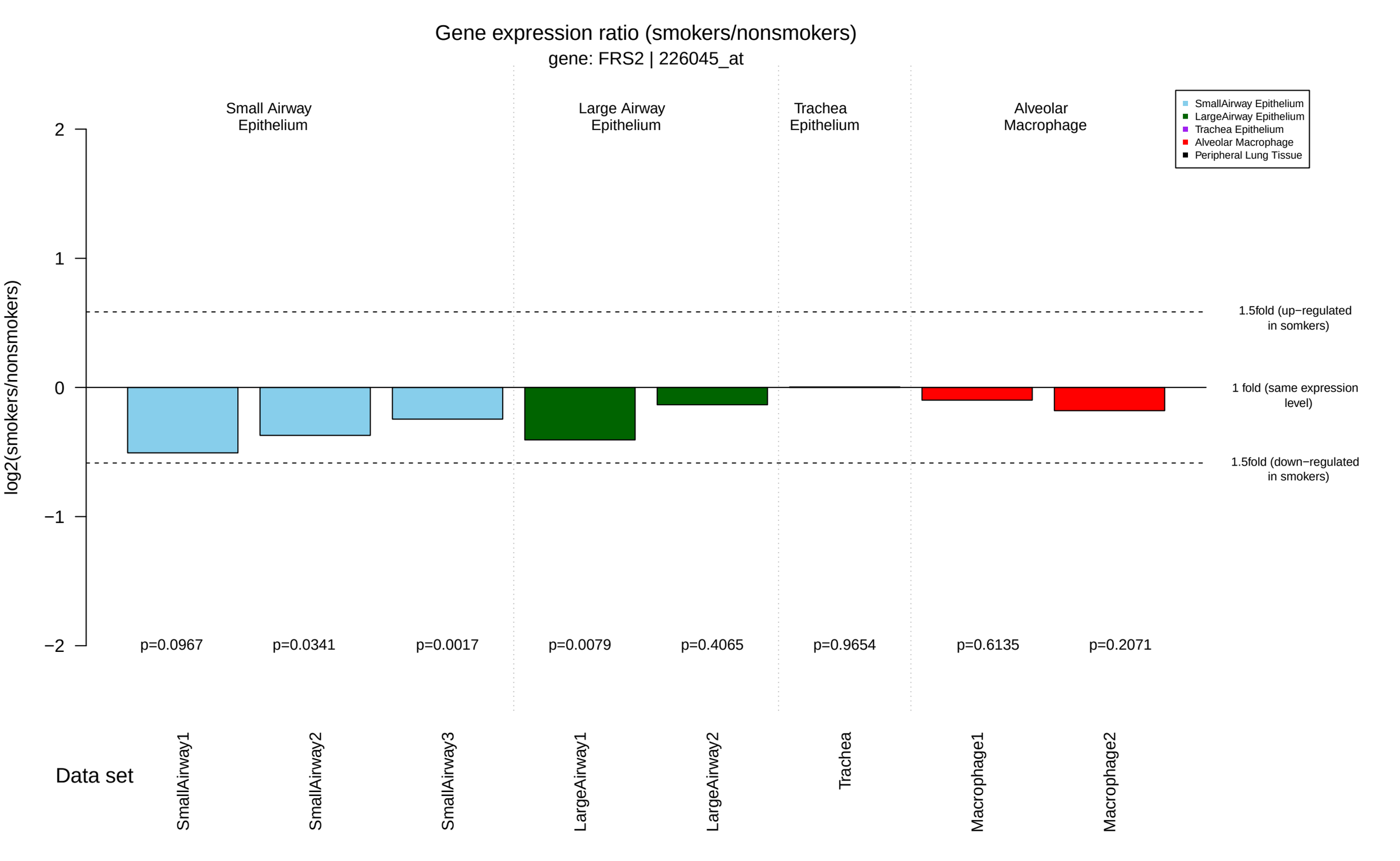
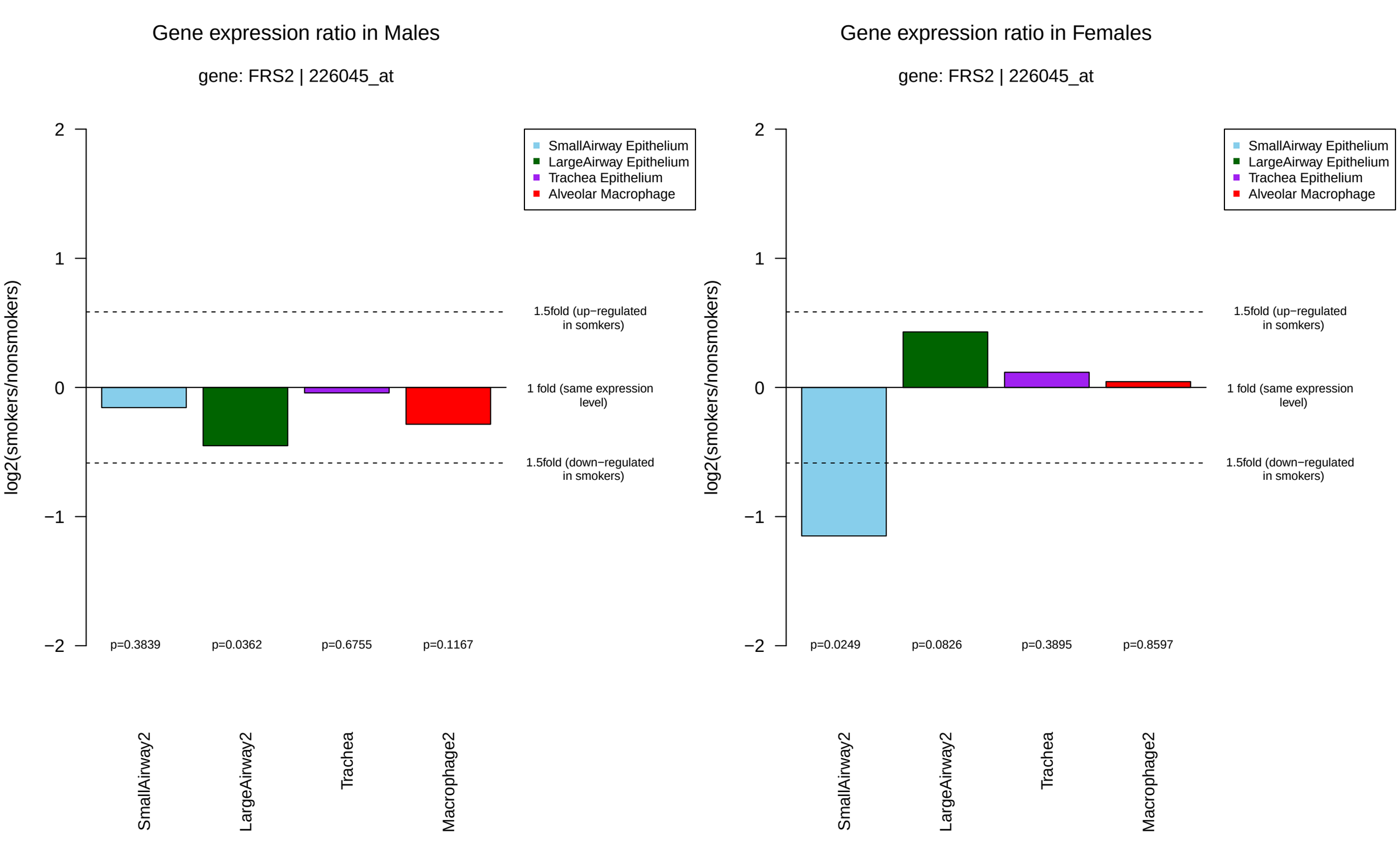
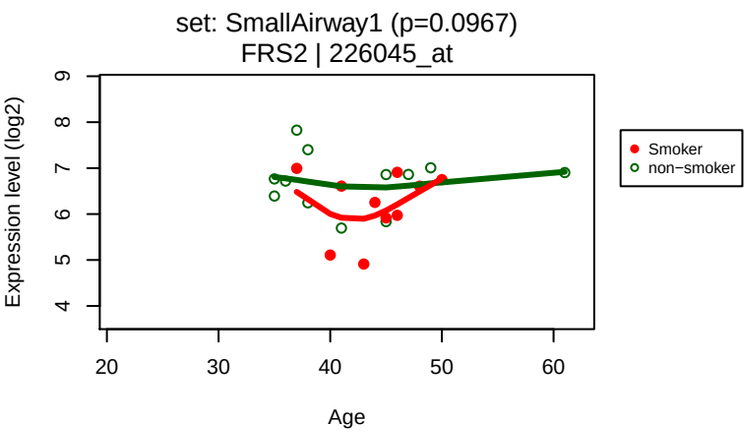
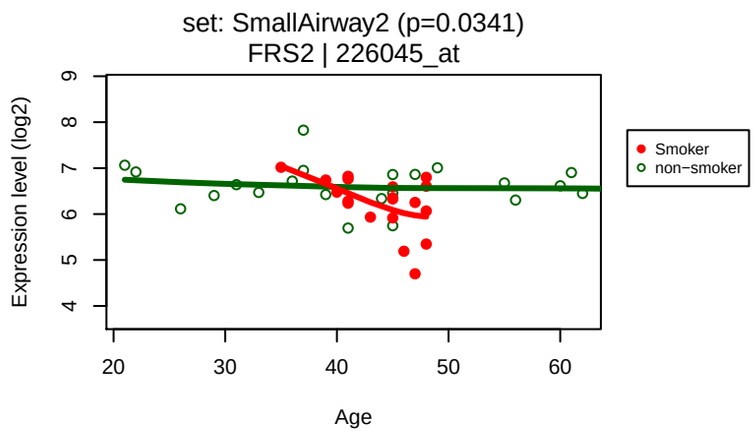
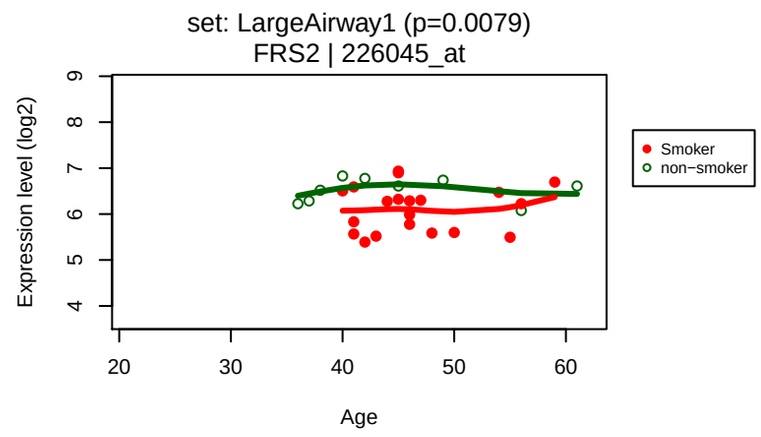

Gene expression of smokers and nonsmokers
gene: FRS2 | 226045_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0967
p=0.0341
p=0.0017
p=0.0079
p=0.4065
p=0.9654
p=0.6135
p=0.2071
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

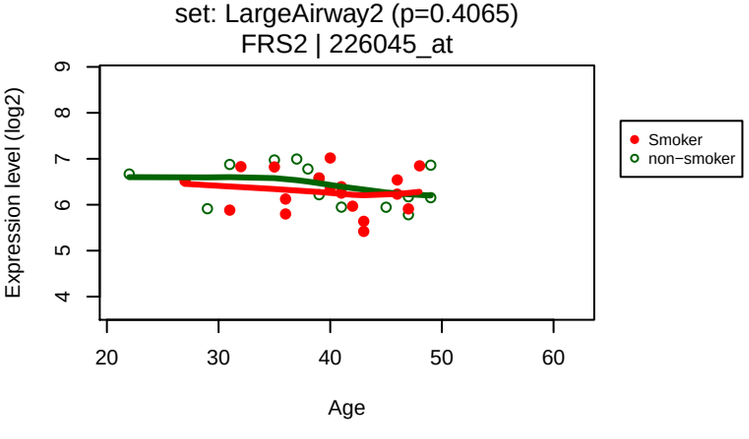

Gene expression ratio (smokers/nonsmokers)
gene: FRS2 | 226045_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0967
p=0.0341
p=0.0017
p=0.0079
p=0.4065
p=0.9654
p=0.6135
p=0.2071
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

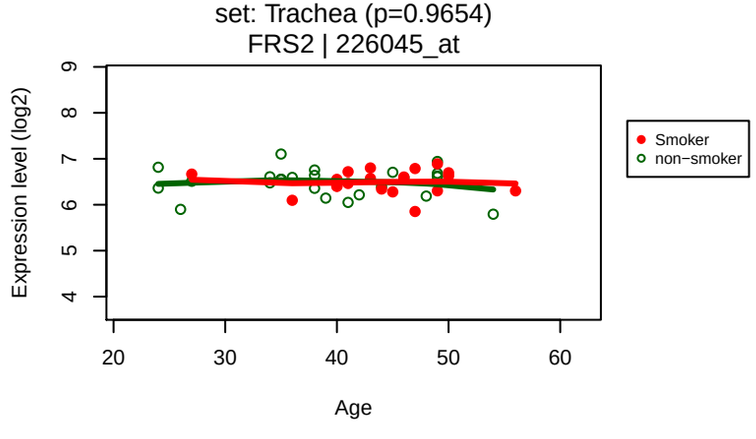

Gene expression ratio in Males
gene: FRS2 | 226045_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3839
p=0.0362
p=0.6755
p=0.1167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRS2 | 226045_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0249
p=0.0826
p=0.3895
p=0.8597
SmallAirway2
LargeAirway2
Trachea
Macrophage2

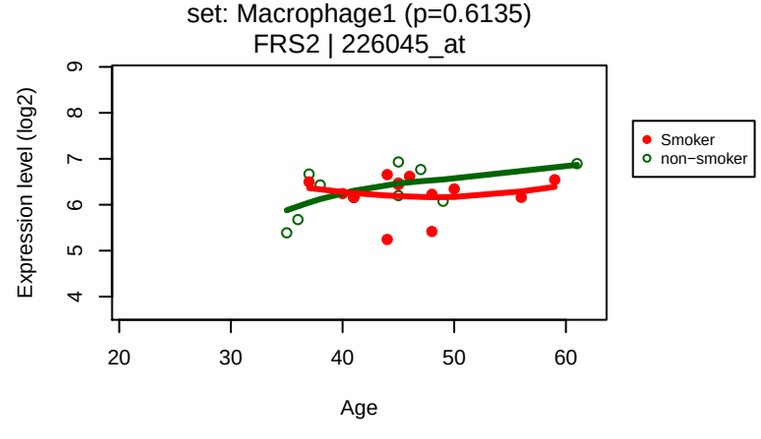

set: SmallAirway1 (p=0.0967)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker

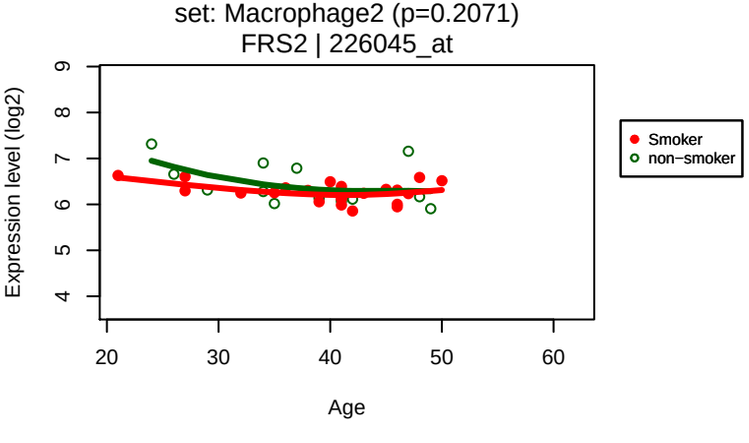

set: SmallAirway2 (p=0.0341)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker

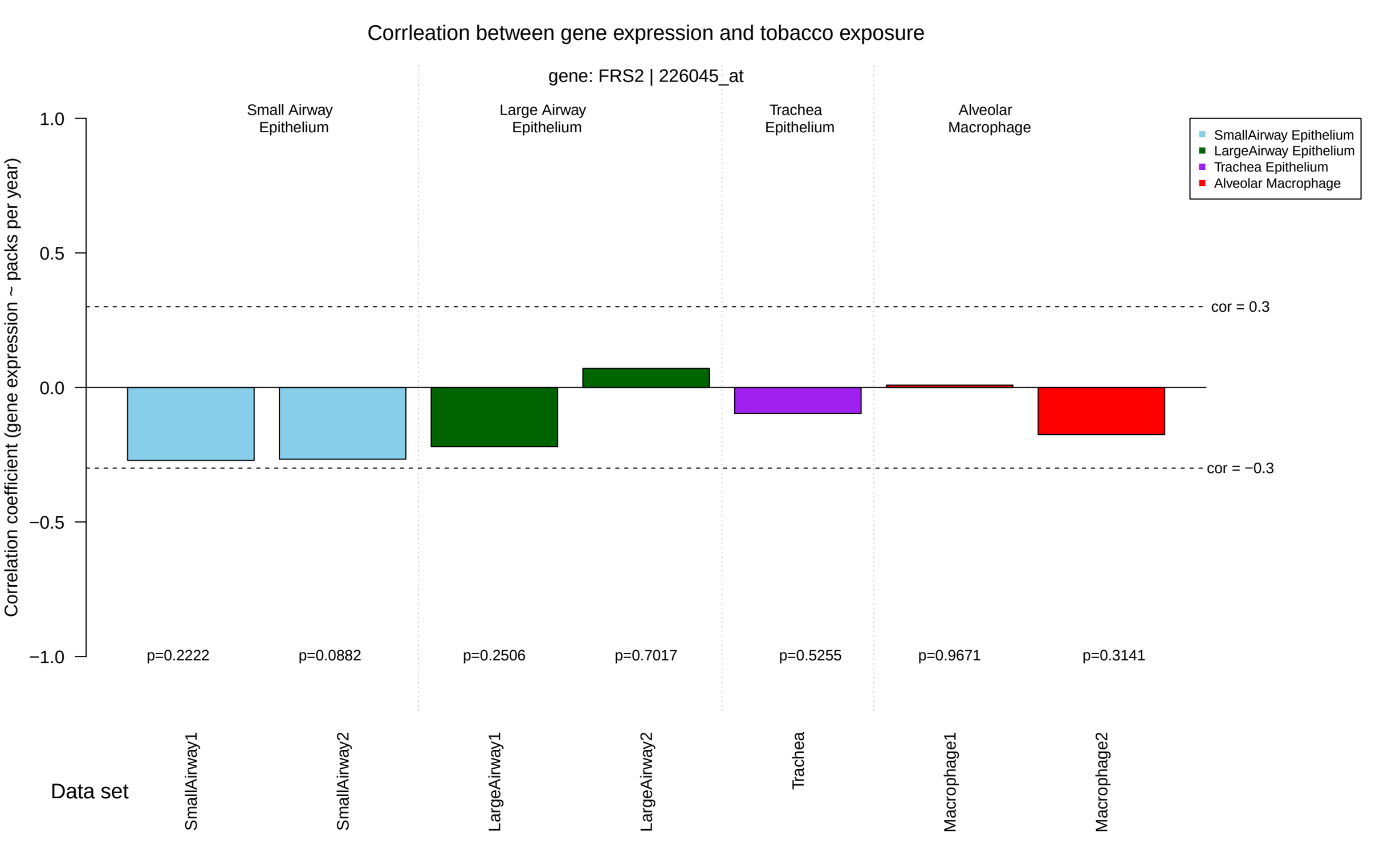

set: LargeAirway1 (p=0.0079)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4065)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9654)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6135)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2071)
FRS2 | 226045_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRS2 | 226045_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2222
p=0.0882
p=0.2506
p=0.7017
p=0.5255
p=0.9671
p=0.3141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2